Dylan Becchetti

Engineering Software & Automation | Mechanical Engineering

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Portfolio: formsby-systems.github.io/engineering-portfolio

ENGINEERING EXPERIENCE

Cardinal Glass Industries | St. Louis Park, MN

Continuous Improvement Engineering Intern, summers 2024-2025 | Engineering Intern, summer 2023

- Built a self-directed Excel/VBA production tracker used by engineers and operators in daily R&D shift debriefs.
- Automated weekly quality counts from a 10,000+ unit log and connected production, yield and work-order reporting in Excel/VBA while preserving existing source worksheets.
- Joined purchasing and parts records in Smartsheet using standardized identifiers and a separately sorted maintenance report; planned storage and signage across 14 locations.
- Wrote illustrated sample-build instructions, measured cleanroom airflow with a hood and micromanometer, analyzed data in Minitab and drafted AutoCAD layouts.

SELECTED PROJECTS

Midas Futures | Creator & Technical Lead

- Built a Python futures research and demo trading platform spanning modular strategies, batch experiments, ablations, Monte Carlo studies and shared-risk account simulation.
- Built checkpointed cloud research and a durable job queue; recovered a 72-case study without rerunning completed cases; tested queue holds for unverifiable claimed runs.

Operalith | Founder & Developer, independent product

- Built a configurable React/Next.js workspace connecting customer enquiries, projects, tasks and supplies through shared records.
- Implemented enquiry-to-project conversion and conflict-aware undo; built PostgreSQL save checks for ownership, stale edits and repeated requests.

Kinetic Wilbur | Mechanical Design, year-long team capstone

- Designed and iterated the full-scale OpenSCAD assembly for an 80-inch suspended planetary-gear sculpture, translating stakeholder needs and sponsor feedback into manufacturable parts.
- Modeled steel carrier arms and cable stays, designed staggered ring layers from four identical CNC half-cuts, and revised drive alignment with team FEA and fit checks.

Formsby | Creator, cross-source search and matching

- Built Python discovery and Next.js/PostgreSQL housing-search prototypes with ranked, source-linked shortlists; removed duplicate URLs and kept partial results and private, owner-scoped search history.

Scanned Leverage and Optimized Production | Creator

- Built Python/Luau market research and modular prototype tools; generated and reproduced two original Roblox arenas from shared game logic, with configuration, route and stale-input checks.

EDUCATION

Lehigh University | B.S. Mechanical Engineering, expected January 2027

Materials Science and Engineering minor | Coursework expected December 2026 | CSWA, December 2024

TECHNICAL SKILLS

Python, JavaScript, React/Next.js, SQL/PostgreSQL, Git, automated testing, OpenSCAD, SOLIDWORKS, AutoCAD, Excel/VBA, Smartsheet, Minitab.